

BARCELONA, SPAIN • ARCHITECTURAL MONOGRAPH



BIMCO

VOLUME III: COMMERCIAL & URBAN MASTERPLANNING

MACRO-SCALE ARCHITECTURAL SYSTEMS & URBAN MORPHOLOGY

Curated Monograph of Commercial Hubs, Retail Atriums, Infrastructure & River Basins

COLLABORATIVE LEADERSHIP & PRINCIPALS

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BIM Director & Computational Architect | Senior Architectural Designer & Visualizer

BIMCO STUDIO • BARCELONA, SPAIN

BARCELONA, SPAIN • EDITION 2026

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SPAIN

THE METROPOLITAN SCALE & URBAN LIFE

Connectivity, Spatial Complexity, and Resilient Public Realms

Architecture reaches its most profound social resonance when operating at the scale of the city. At BIMCO Studio Barcelona, commercial and infrastructure complexes are not isolated objects; they are vital civic anchors that channel urban movement, stimulate commerce, and generate social encounters.

From the industrial wholesale arteries of the Shargh Steel Complex to the fluid retail gallerias of the Erbil Department Store, and the regenerative ecology of the Zargandeh river valley, our urban masterplans synthesize complex logistic flows with humane public realms. Every project is grounded in rigorous BIM and engineering coordination.

COMMERCIAL & URBAN INDEX

Structured Progression from Logistics Complexes to Civic Masterplans

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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

From Barcelona, BIMCO approaches urban planning through systemic integration of infra...

BIM & STRUCTURAL INTEGRITY SPECS

GIS Spatial Analytics, Large-Span Steel Truss BIM LOD 350, Parametric Shells, Mi...



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Mall design focuses on functionality, aesthetics, and customer experience by integrating a logical layout with amenities like clear pathways and ample parking, while also incorporating elements such as natural light, interactive digital displays, and diverse entertainment and dining options to create a vibrant, multi-use destination

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Integrated industrial-commercial campus combining wholesale steel retail, bank branch...

BIM & STRUCTURAL INTEGRITY SPECS

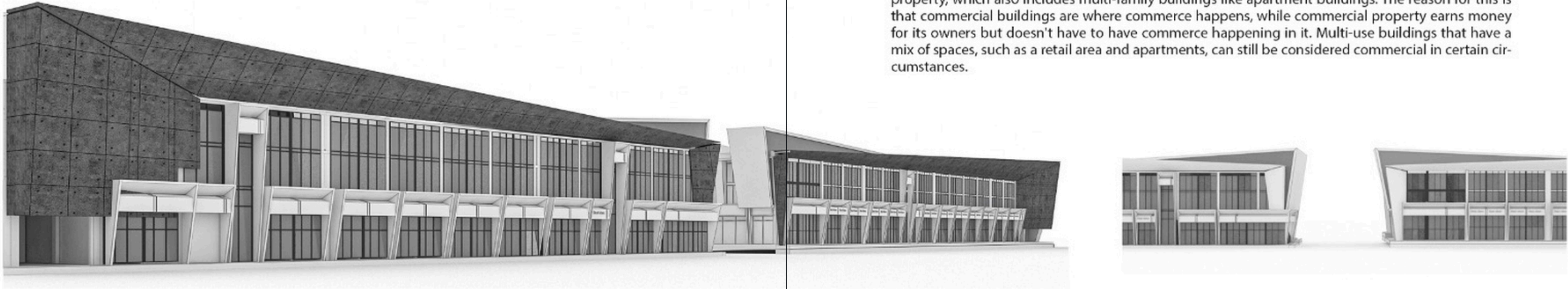
Heavy vehicular circulation turning radii, 36m clear-span steel trusses, fire se...

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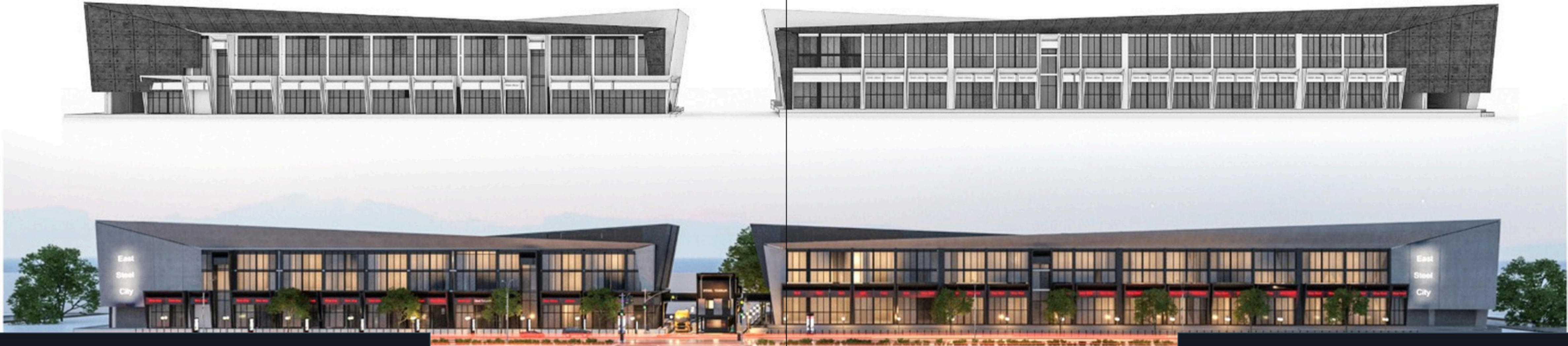
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Commercial buildings are buildings where commercial activities take place. Commercial buildings include office buildings, retail space, warehouses and more. This differs somewhat from commercial property, which also includes multi-family buildings like apartment buildings. The reason for this is that commercial buildings are where commerce happens, while commercial property earns money for its owners but doesn't have to have commerce happening in it. Multi-use buildings that have a mix of spaces, such as a retail area and apartments, can still be considered commercial in certain circumstances.



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Dual-function commercial facility combining high-bay inventory storage with glazed di...

BIM & STRUCTURAL INTEGRITY SPECS

15-ton overhead gantry crane coordination, insulated sandwich panels, heavy-duty...



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Commercial buildings are an important part of doing business, both for those who own them and for those who rent them. A commercial building could contain one or more of many types of businesses, including retailers, restaurants, offices or manufacturing. If you are planning to open a business with a physical location or you want to own and manage commercial properties, it's important to understand what this involves.



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Corporate headquarters and full-service commercial bank branch featuring dynamic angl...

BIM & STRUCTURAL INTEGRITY SPECS

Anthracite aluminum composite rainscreens, point-supported glass fittings, centr...

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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Executive street elevation with synchronized loading docks, automated rolling section...

a strong metal that is a mixture of iron and carbon, used for making things that need a strong structure, especially vehicles and buildings:
steel girders/rods/struts
a steel helmet
a steel-plated army truck

More examples
a steel mill
Coke is used in the production of steel.
The concrete is strengthened with steel rods.
Guitar strings nowadays are made from steel or nylon.
The gates were locked with a padlock and a heavy steel



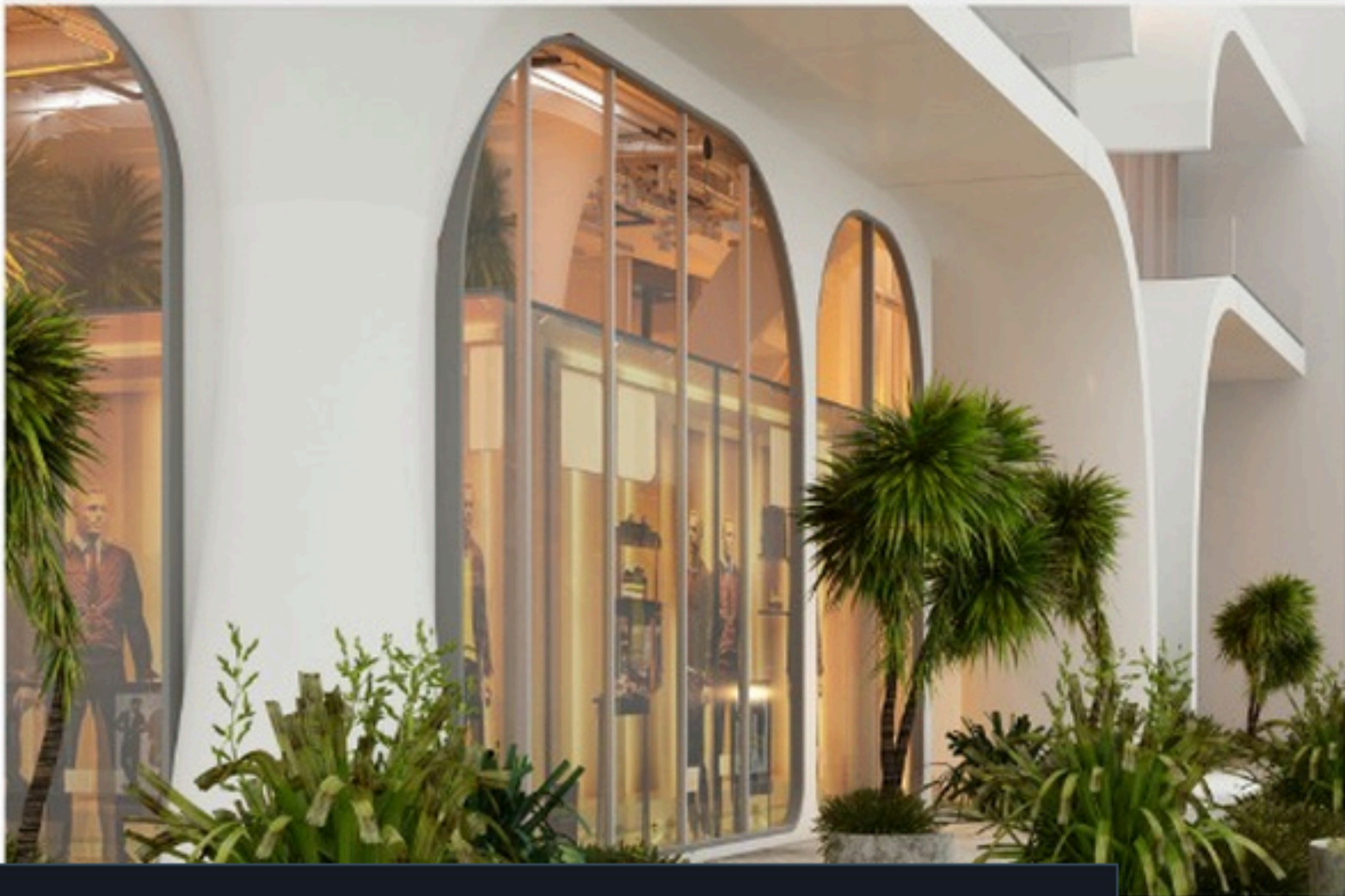
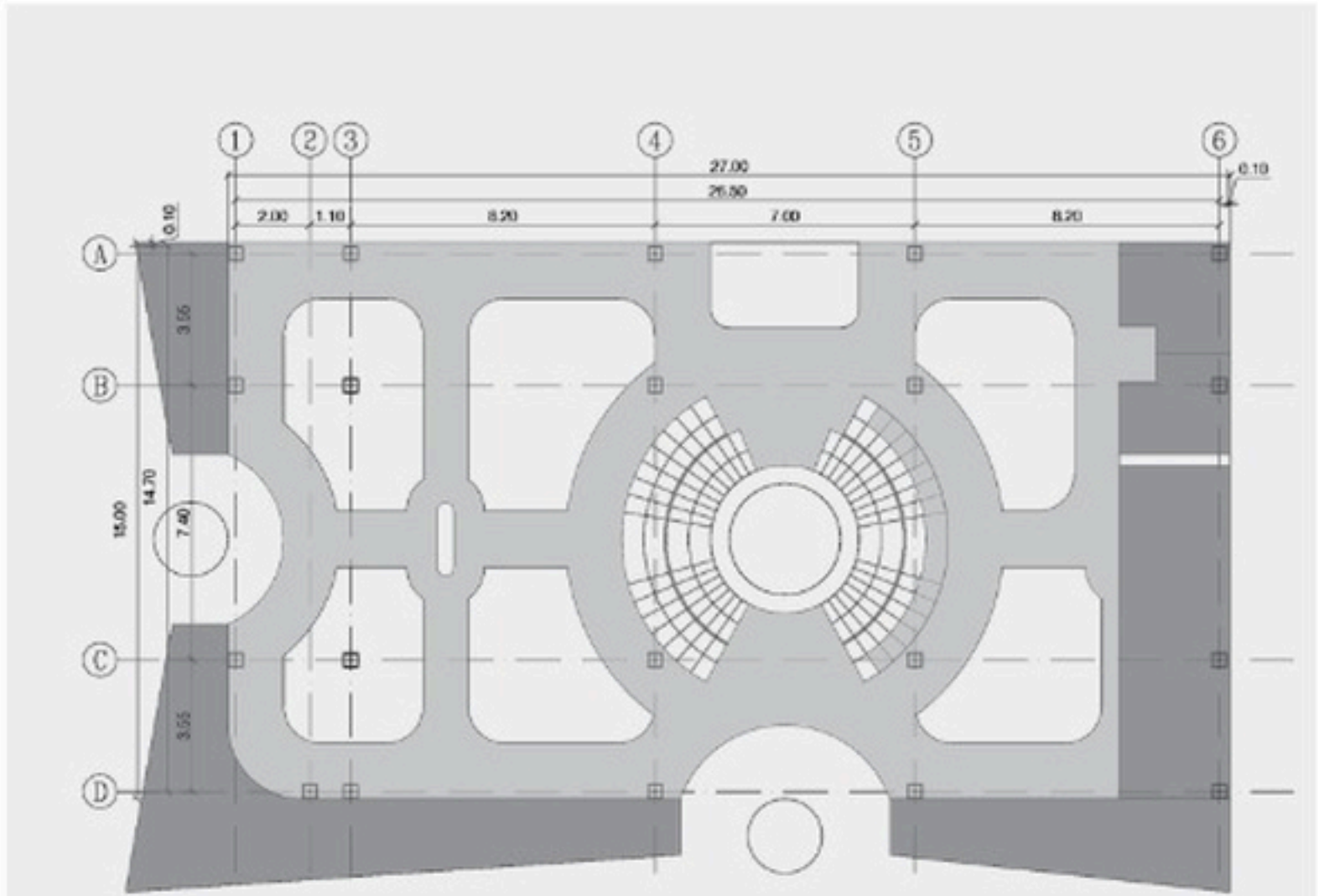
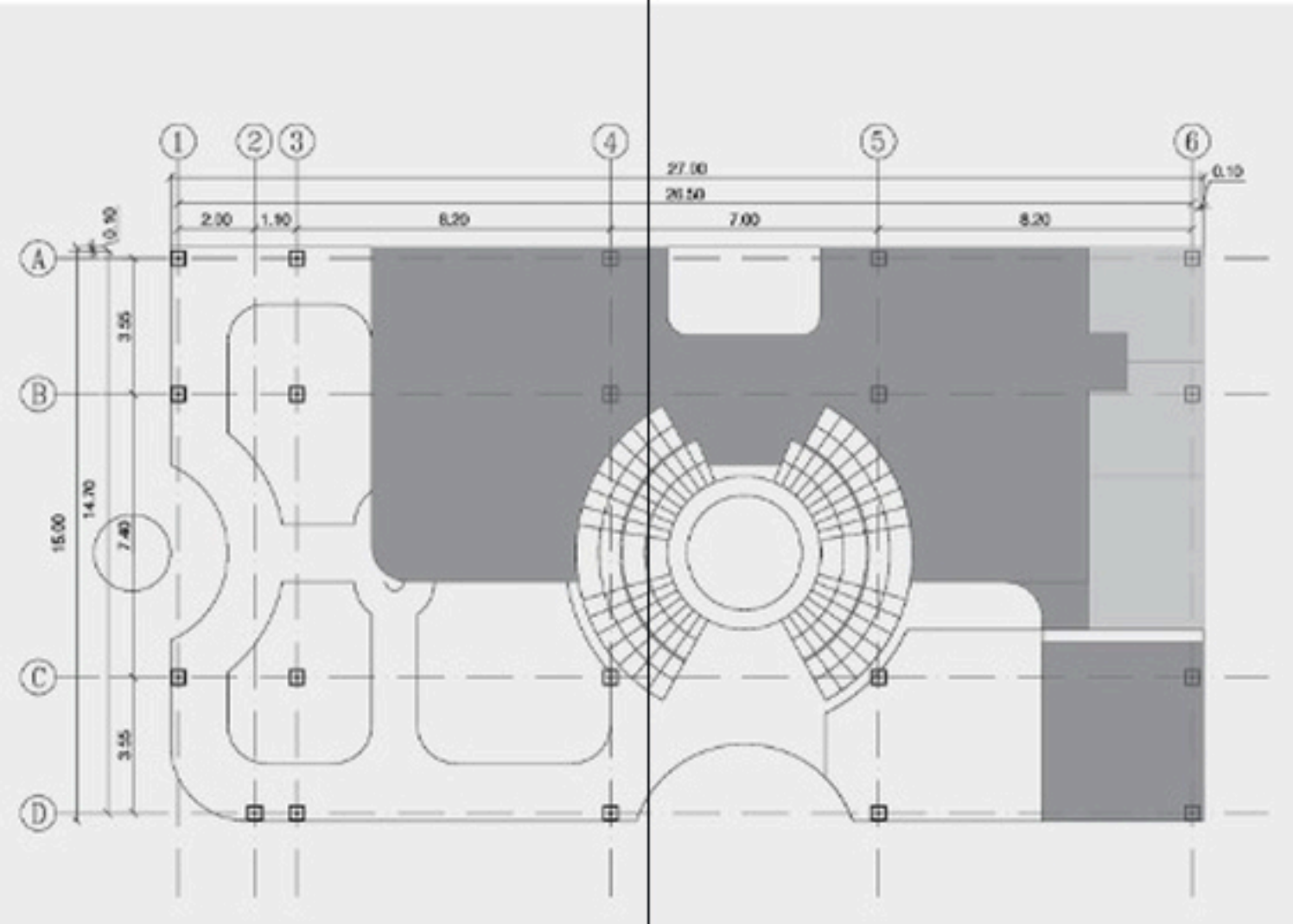
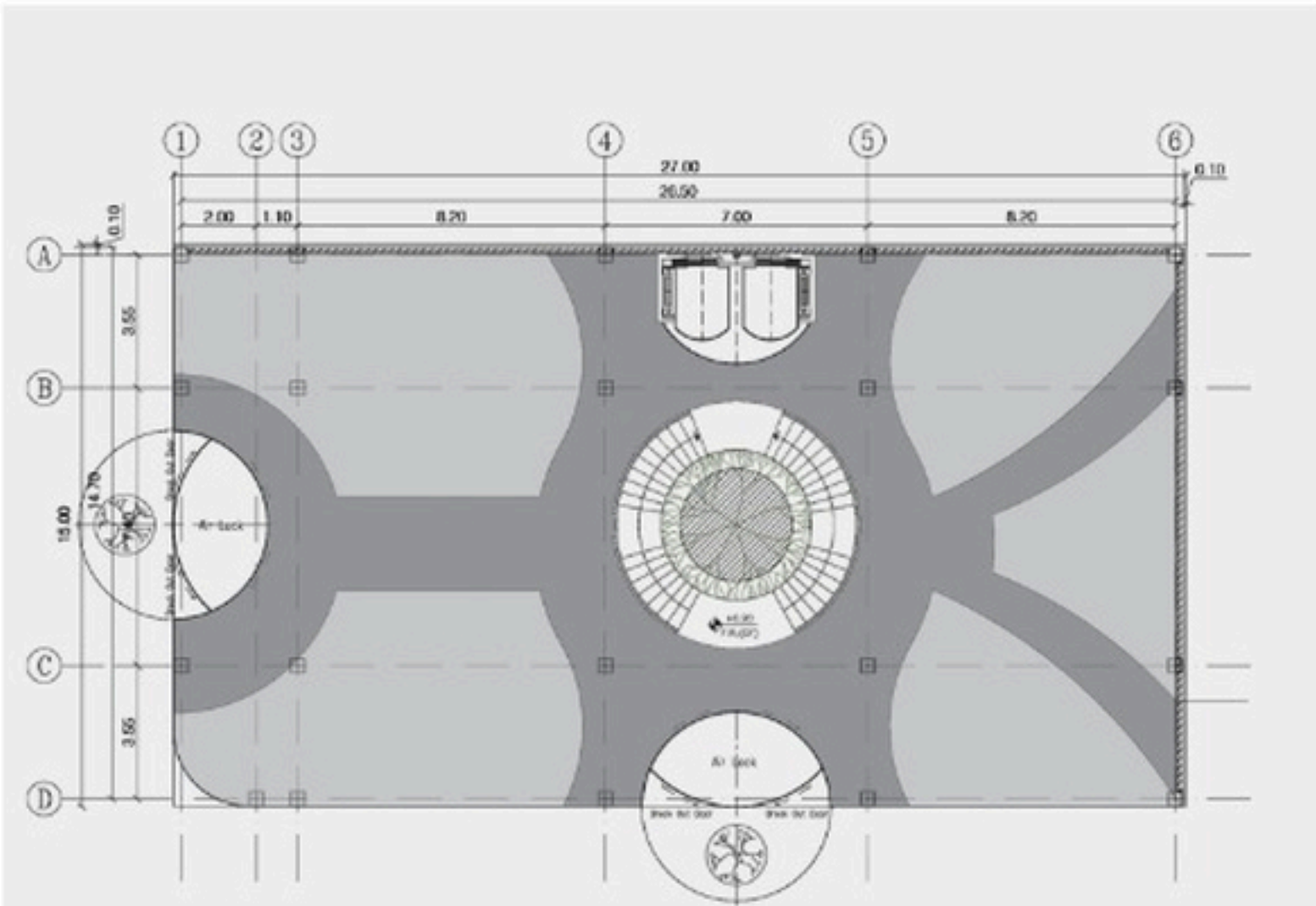
BIM & STRUCTURAL INTEGRITY SPECS

Hydraulic dock levelers, semi-trailer circulation clearance, architectural exter...

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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Regional department store landmark in Erbil featuring monumental parabolic arches and...

BIM & STRUCTURAL INTEGRITY SPECS

Parametric curved steel arches, exterior tensioned fabric solar screens, high-ca...

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Mainline department store or simply, the traditional department store, offering mid-to high-end goods, most or at least some .of the time at the full retail price



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Expansive retail galleria centered around an elliptical daylight atrium with panorami...

BIM & STRUCTURAL INTEGRITY SPECS

Atrium smoke extraction venting, automated NFPA sprinkler zoning, acoustic baffl...

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ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Comprehensive retail floor plans detailing anchor department zones, boutique tenant L...



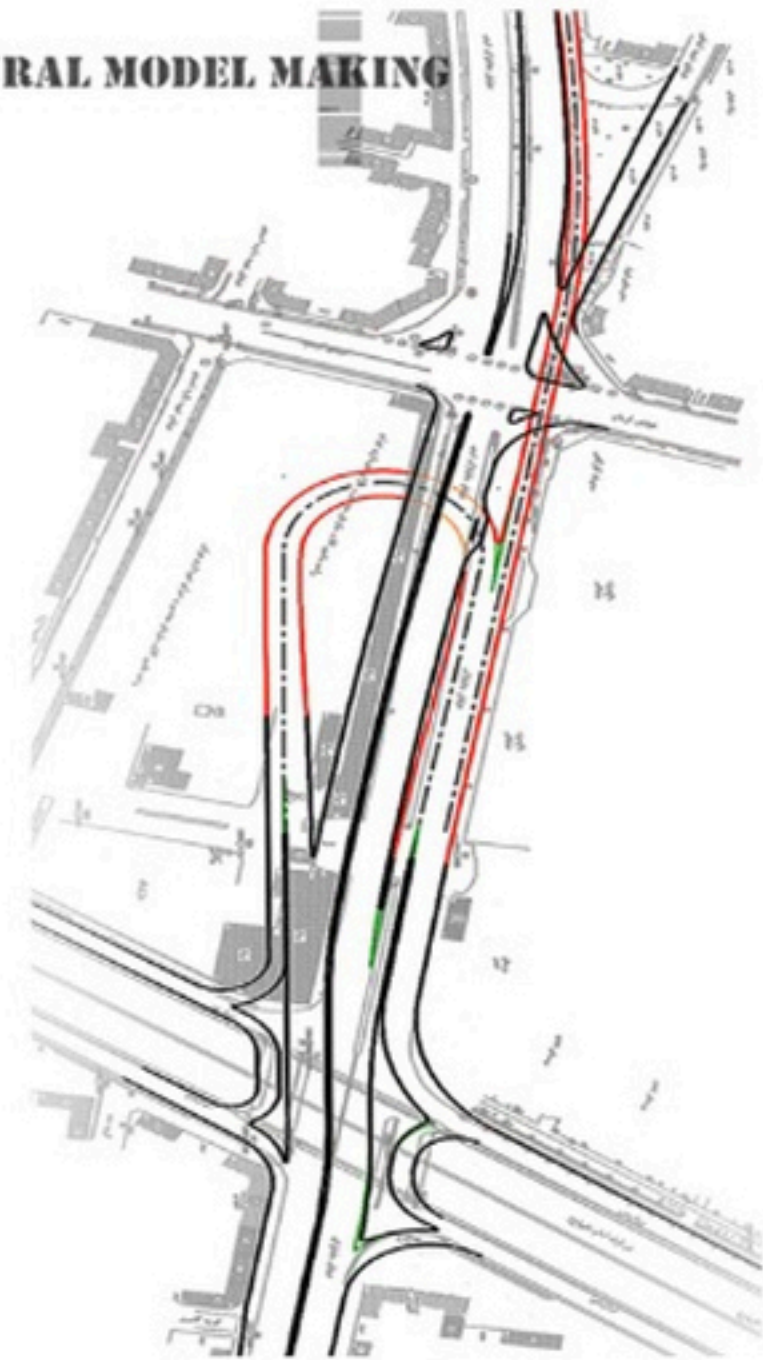
departments often include the following: clothing, cosmetics, do it yourself, furniture, gardening, hardware, home appliances, houseware, paint, sporting goods, toiletries, and toys. Additionally, other lines of products such as food, books, jewellery, electronics, stationery, photographic equipment, baby products, and products for pets are sometimes included

BIM & STRUCTURAL INTEGRITY SPECS

Post-tensioned two-way slab (12x12m column grid), emergency egress stair calcula...



ARCHITECTURAL MODEL MAKING YEAR 2014

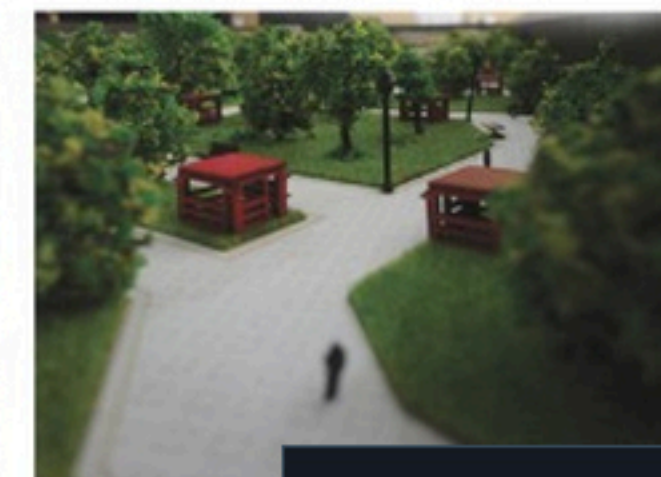


It is also key to remember that models are just as much for the architect and the firm as they are for external stakeholders and clients. As explained in the Architect's Journal, models are actually a vital part of the creative process. When a designer is putting together a model, it gives them a chance to see their vision in miniature and give them space to consider what could be improved in real-time.



ARCHITECTURAL MODEL MAKING YEAR 2014

Perhaps the main reasons why architectural models are so important is because they can highlight the features and utility of a planned structure in a way that a composite sketch or computer-generated image could never do. If you want to demonstrate how the windows of a building will allow light to filter through a space, a model is an ideal way to do so.



ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Civil and urban highway interchange optimizing vehicle flow through multi-level curve...

BIM & STRUCTURAL INTEGRITY SPECS

Prestressed concrete box-girder superstructures, elastomeric bridge bearings, st...



Physical Organize on Urban Fabric with an Organic Point of View on Morphology

Urban spaces have been considered as urban life cells since long ago and play a basic role in urbanization. In this space is not long ago different events such as sports, religious and entertainment had displayed but now because of urbanization improvement and cities development they lost their importance. This makes some problems such as environmental pollution, problems in traveling, security paths, degeneration of historic center and quality drop in urban spaces. Urban space role as a place for social relations and also a place for providing social culture is disappearing gradually in our country. Dominant conditions in cities offer a modern life style. For example shopping urban space designing for more vitality space. Automobiles are now dominant in urban spaces and building paths and urban spaces are now far away from "walker" needs and determinants, so social and cultural values and attractions decrease. In such a progress context and application of urban factors such as neighborhood, street, square, and public domain and... changed and lost its human content. According to physical and functional aspects of urban spaces and social and cultural features as well as social behaviors and relations and also evaluation of conceptual features which are affected by physical aspects are some important concepts that paying attention to them can effect on vitality of urban spaces. Different views toward vitality criteria and its concept in urban designing since have been 1981 evaluated. Jean Jacobs in "Death and Life in American Cities" described vitality in streets as a chance for having good relations between people. Social places provide comfortable and meeting setting with good lights. It is necessary to promote vitality of the city in order to achieve vitality. One of vibrant and dynamic urban spaces in the world is the "pedestrian mall" in Iran. Places is not well defined, additionally the large car-dependent thoroughfares and streets in which motor vehicles disturb the city's safety have decreased the city vitality and viability. On the other hand the role of urban public space in



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Case Study: Zargandeh Riverside, Tehran



A part of Design & presentation 2015

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Spatial morphology research evaluating historic development patterns, steep slope ris...

BIM & STRUCTURAL INTEGRITY SPECS

GIS spatial statistics, parcel density mapping, slope stability calculations, ri...

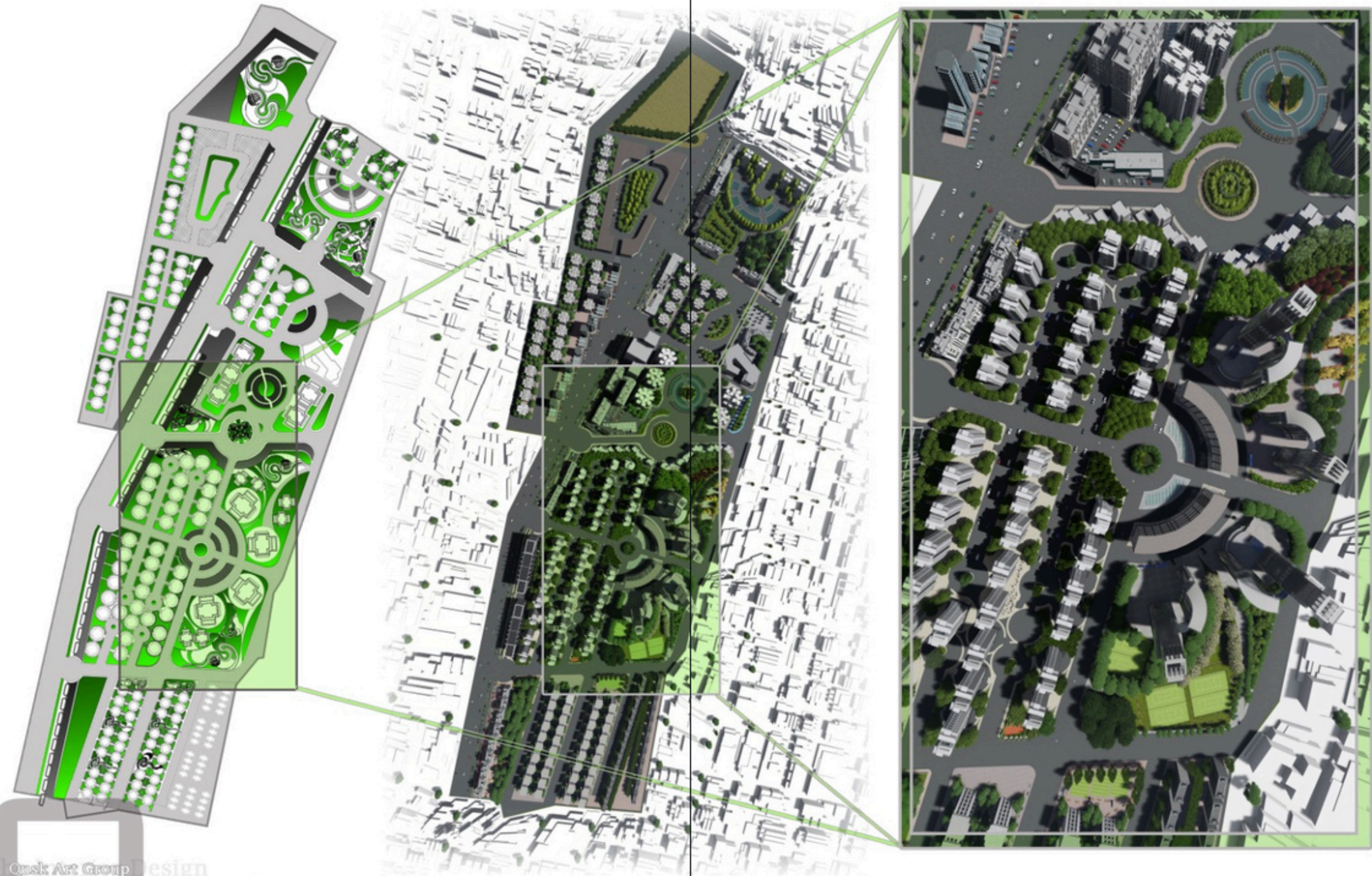


Architecture Company Profile

Bioengineered slope stabilization, permeable surfaces, urban bioswales and droug...



urban plan design



Onsk Art Group Design
Architecture Company Profile

A part of Design & presentation 2018

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Macro urban massing strategy sculpting high-rise envelopes to maximize street-level s...

BIM & STRUCTURAL INTEGRITY SPECS

Ladybug daylight envelope analysis, CFD urban microclimate modeling, Floor Area ...



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Urban Design



A part of Design & presentation 2018

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

Transit-oriented district masterplan integrating high-speed rail, metro stations, ped...

BIM & STRUCTURAL INTEGRITY SPECS

Pedestrian isochrone 5/10 min walksheds, underground automated logistics, distri...

BIMCO PROTOCOL: DESIGN & MODELING

Stages 01 – 03 • Urban & Commercial Protocol

STAGE 1: GIS & MACRO-SCALE MOBILITY

Spatial network analytics, isochrone pedestrian walkshed mapping, traffic flow modeling.

STAGE 2: URBAN MASSING & SOLAR ENVELOPE

Ladybug solar rights study, sky exposure planes, wind tunnel CFD microclimate optimization.

STAGE 3: MULTI-SYSTEM BIM LOD 350

Heavy steel clear-span trusses, long-span post-tensioned slabs, subterranean utilities.

BIMCO PROTOCOL: VISUALIZATION & BUILD

Stages 04 – 06 • From CGI to On-Site Reality

STAGE 4: DAYLIGHT ATRIUM & NIGHT SKYLINE

Corona high-bay photometric simulations, nocturnal retail visibility, material mockups.

STAGE 5: FULL CIVIC & RETAIL DRAWING SET

1:50 concourse layouts, emergency egress stairs, dock loading clearances, MEP shafts.

STAGE 6: PHASING, MOBILITY & SITE AUDIT

Construction staging schedules, precast girder crane sequence, site quality assurance.

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

BIMCO Studio Barcelona mandates complete continuity from parametric BIM LOD 350 to si...

BIM & STRUCTURAL INTEGRITY SPECS

ISO 19650 BIM Standards, Autodesk Revit LOD 350, Clash Zero Tolerance, Worksite ...

SOHEIL MASTI

BIM Director & Computational Architecture Lead

BIMCO Studio Barcelona • Spain & International

BIM MANAGEMENT & INFORMATION ARCHITECTURE:

- Strategic BIM implementation, Revit LOD 350 parametric modeling
- Multi-discipline clash audits between architectural form, structural frames, and MEP systems
- Automated quantity takeoffs and material schedules for bespoke builds
- International project delivery across Spain, Europe & Middle East
- Digital construction workflows ensuring zero discrepancies on site

SIAVASH PAZOOKI

Senior Architectural Designer & High-End 3D Visualizer

M.Arch & B.Arch • BIMCO Studio Barcelona Partner

CONCEPT DESIGN & VISUALIZATION LEADERSHIP:

- Lead Architectural Concept Designer for signature luxury commissions & masterplans
- Photorealistic architectural rendering, natural lighting moods & cinematic camera angles
- Topographical terrain modeling, landscape synthesis, and material detailing
- Turnkey executive shop drawings, joinery specifications & on-site build supervision
- 8+ years leading award-winning commissions across residential & urban typologies

ARCHITECTURAL BRIEF & SPATIAL CONCEPT

BIMCO Studio Barcelona unites high-level BIM information architecture with signature ...

BIM & STRUCTURAL INTEGRITY SPECS

Registered Architects, BIM Managers (Autodesk Certified), LOD 350 Audit Speciali...



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VOLUME III: COMMERCIAL & URBAN MASTERPLANNING

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